

Section A · Property & Casualty

1. P&C · Multiple choice · Key metric

What does the *Combined Ratio* measure in property and casualty insurance, and what is the correct formula?

A. It measures the profitability of the investment portfolio.

$$\text{CR} = \frac{\text{Financial result}}{\text{Earned premiums}}.$$

B. It measures the technical efficiency of the insurance business (excluding financial results).

$$\begin{aligned}\text{CR} &= \text{Loss Ratio} + \text{Expense Ratio} \\ &= \frac{\text{Incurred losses}}{\text{Earned premiums}} + \frac{\text{Total expenses}}{\text{Earned premiums}}.\end{aligned}$$

A CR < 100% indicates an underwriting profit.

C. It measures the available solvency margin relative to the required margin.

$$\text{CR} = \frac{\text{Technical capital}}{\text{Minimum solvency margin}}.$$

D. It measures the gross loss ratio.

$$\text{CR} = \frac{\text{Paid losses}}{\text{Written premiums}}.$$

Question 1: Correct answer: B

The **Combined Ratio (CR)** is the central technical metric in P&C insurance. It measures the percentage of each premium unit used to pay losses and operating expenses, *excluding financial results*.

$$\text{CR} = \text{Loss Ratio} + \text{Expense Ratio}$$

$$\text{Loss Ratio} = \frac{\text{Incurred losses}}{\text{Earned premiums}}$$

The *Loss Ratio* indicates the percentage allocated to losses.

$$\text{Expense Ratio} = \frac{\text{Total expenses}}{\text{Earned premiums}}$$

The *Expense Ratio* indicates the percentage allocated to expenses.

- CR < 100%: **Underwriting profit** (the insurer earns a profit before investment income).
- CR = 100%: **Underwriting break-even point**.
- CR > 100%: **Underwriting loss** (profitability depends on financial results).

In developed markets, a CR of 95–98% is common and manageable because of investment income. A sustained CR > 105% usually indicates structural pricing or risk-selection problems.

Option D confuses the CR with the simple *gross Loss Ratio*; option A confuses it with investment indicators. These are common interview mistakes, and understanding the distinction differentiates a senior candidate.

2. P&C · Property Multiple choice · Underwriting

In property insurance underwriting, what is the difference between *PML* (Probable Maximum Loss) and *MFL* (Maximum Foreseeable Loss)?

- A. They are equivalent and interchangeable terms. Both represent 100% of the insured value.
- B. PML is the maximum loss assuming that protection systems (sprinklers, fire brigades) *operate correctly*; MFL is the maximum loss assuming that *all systems fail simultaneously*. Always $MFL \geq PML$. The difference determines the reinsurance program.
- C. PML applies to industrial risks; MFL applies exclusively to smaller commercial risks.
- D. PML measures the probable loss over one year; MFL measures the accumulated loss over ten years under normal operating conditions.

Question 2: Correct answer: B

The PML/MFL distinction is fundamental when underwriting large industrial risks (petrochemical plants, refineries, and power plants):

- **PML (Probable Maximum Loss):** an estimate of the maximum loss assuming that fire-protection systems (automatic sprinklers, firewalls, and the internal fire brigade) *operate as designed*. It may equal 40–70% of the insured value in well-protected facilities.
- **MFL (Maximum Foreseeable Loss):** an estimate of the maximum loss assuming that *all protection systems fail simultaneously* in the worst credible scenario. It may reach 80–100% of the insured value. It is the “worst credible case.”

The relationship is **always** $MFL \geq PML$. The underwriter uses the PML to define the insurer’s retention limit and the facultative reinsurance program. If the PML exceeds retention capacity, the difference up to the MFL is ceded to reinsurance to protect the balance sheet against a catastrophe.

HDI/Talanx tip: in a group that also controls Hannover Re, designing a PML/MFL-based reinsurance program is a first-rate technical matter. A senior actuary who masters it adds direct value.

3. P&C Calculation · Combined Ratio and underwriting result

Using the following 2025 data for HDI Colombia's property portfolio, calculate the *Loss Ratio*, *Expense Ratio*, and *Combined Ratio*, and determine whether the business was technically profitable.

Item	COP million
Net earned premiums	8,500
Net incurred losses (paid + IBNR)	5,440
Commissions and acquisition expenses	1,020
Administrative expenses allocated to the line	595

Formulas

$$\begin{aligned}\text{Loss Ratio} &= \frac{\text{Incurred losses}}{\text{Earned premiums}} \\ \text{Expense Ratio} &= \frac{\text{Acquisition expenses} + \text{Administrative expenses}}{\text{Earned premiums}} \\ \text{Combined Ratio} &= \text{Loss Ratio} + \text{Expense Ratio} \\ \text{Underwriting result} &= \text{Earned premiums} - \text{Losses} \\ &\quad - \text{Total expenses}\end{aligned}$$

What does a CR at this level mean for the actuary's pricing strategy?

Question 3: CR = 83.0% — Technically profitable business

$$\begin{aligned}\text{Loss Ratio} &= \frac{5,440}{8,500} = \mathbf{64.0\%} \\ \text{Expense Ratio} &= \frac{1,020 + 595}{8,500} = \frac{1,615}{8,500} = \mathbf{19.0\%} \\ \text{Combined Ratio} &= 64.0\% + 19.0\% = \mathbf{83.0\%} \\ \text{Underwriting result} &= 8,500 - 5,440 - 1,615 \\ &= \mathbf{1,445 \text{ COP million}} \quad (\text{underwriting profit})\end{aligned}$$

A **CR of 83.0%** is excellent for a property portfolio: for each peso of premium, 64 cents go to losses, 19 cents to expenses, and **17 cents remain as underwriting margin** before financial results.

Implication for the actuary: a CR well below 100% may indicate that rates are excessively conservative (creating a risk of losing volume to more aggressive competitors) or that the period

experienced favorable loss activity (good years within market cycles). The senior actuary must assess whether the result is sustainable or reflects unusually low loss activity, verify that IBNR is properly estimated, and determine whether the portfolio has catastrophe exposure not captured in historical data.

4. Liability Multiple choice · Coverage basis

What is the fundamental difference between a *Claims Made* liability policy and an *Occurrence* policy?

- A. *Claims Made*: covers losses occurring during the policy period, regardless of when they are reported. *Occurrence*: covers losses reported during the policy period, regardless of when they occurred.
- B. *Claims Made*: covers claims *made* during the policy period (potentially subject to a retroactive date). *Occurrence*: covers events that *occur* during the policy period, regardless of when the claim is made—even years later. Actuarial impact: Occurrence coverage generates very long-tail IBNR.
- C. They are actuarially equivalent; the difference is purely contractual and does not affect the IBNR reserve.
- D. *Claims Made* applies only to professional liability; *Occurrence* applies only to product liability and employers' liability.

Context: In Colombia, Directors and Officers (D&O), medical professional liability, and engineering professional liability are typically written on a *Claims Made* basis. Motor liability (SOAT) and product liability are typically structured on an *Occurrence* basis.

Answer: B

This distinction is critical in liability insurance and is one of the most common questions in P&C actuarial interviews:

- **OCCURRENCE:** The policy responds when the **EVENT** occurs during the policy period, regardless of when the claim is made (possibly ten years later). It generates very long-tail IBNR.
- **CLAIMS MADE:** The policy responds when the **CLAIM** is made during the policy period. The insurer may define a retroactive date from which events are accepted. Exposure closes when the policy ends (unless run-off coverage exists), making IBNR more manageable.

Actuarial impact: Occurrence liability policies may have development tails of 10–20 years, especially for product, environmental, and construction liability. This makes IBNR the most significant

and difficult reserve to estimate. Claims Made policies transfer part of the tail risk to the insured, who must maintain continuous coverage, or to the run-off market.

Specific example: a building constructed in 2010 develops a structural defect discovered in 2025. The contractor's Occurrence liability policy in force in 2010 could still respond; a Claims Made policy without continuous renewal would not.

5. Liability Calculation · Per-occurrence and aggregate limits

A company has a General Liability policy with the following terms. Three losses occurred during 2025. Calculate how much the insurer pays for each loss and in total, and how much remains the insured's responsibility.

Policy terms:

- Per-occurrence limit: **COP 500 million.**
- Annual aggregate limit: **COP 1,200 million.**
- Deductible: **none.**

Loss	Month	Total claimed loss (COP million)
S-1	March	350
S-2	July	750
S-3	November	480
Total claimed	—	1,580

- How much does the insurer pay for each loss?
- What is the insurer's total indemnity payment?
- What amount remains uninsured and payable by the insured?

Question 5: Insurer pays 1,200 million; insured absorbs 380 million

(a) Loss-by-loss calculation

S-1 (March): Claimed: 350 million. Per-occurrence limit: 500 million. Because $350 < 500$, the insurer **pays 350 million.**

Aggregate used = 350 million,
 Remaining aggregate available = $1,200 - 350 = 850$ million.

S-2 (July): Claimed: 750 million. Per-occurrence limit: 500 million.

$$\min(750, 500) = 500 \text{ million} \quad (\text{per-occurrence limit exhausted}).$$

The insurer **pays 500 million**, and the insured **absorbs 250 million** above the per-occurrence limit.

$$\text{Aggregate used} = 350 + 500 = 850 \text{ million},$$

$$\text{Remaining aggregate} = 1,200 - 850 = 350 \text{ million}.$$

S-3 (November): Claimed: 480 million. The per-occurrence calculation is $\min(480, 500) = 480$ million, which is within the limit. However, only 350 million remains in the aggregate:

$$\min(480, 350) = 350 \text{ million} \quad (\text{aggregate limit exhausted}).$$

The insurer **pays 350 million**, and the insured **absorbs 130 million** above the remaining aggregate.

$$\text{Aggregate used} = 850 + 350 = \mathbf{1,200} \text{ million}.$$

The annual limit is exhausted.

(b) Total paid by the insurer

$$350 + 500 + 350 = \mathbf{1,200} \text{ COP million}.$$

The aggregate limit is exactly exhausted.

(c) Total uninsured amount payable by the insured

$$\text{Total claimed} = 1,580 \text{ COP million},$$

$$\text{Total covered} = 1,200 \text{ COP million},$$

$$\text{Amount payable by the insured} = 1,580 - 1,200$$

$$= \mathbf{380} \text{ COP million}.$$

Breakdown: 250 million above the S-2 occurrence limit plus 130 million caused by exhaustion of the aggregate in S-3.

Actuarial lesson: this exercise illustrates why the actuary must assess limit adequacy during pricing. If expected loss activity may exceed the aggregate limit, the insured remains underinsured. The actuary should recommend higher limits or assess whether the current premium adequately reflects the risk up to the contracted limit.

6. Liability · Colombia Multiple choice · SOAT

Which of the following statements about Colombia's SOAT (Compulsory Traffic Accident Insurance) is correct?

- A. SOAT is voluntary liability insurance; the insured must prove the driver's fault to access benefits.
- B. SOAT is compulsory for every motor vehicle operating in Colombia. It covers accident *victims*, regardless of fault, through health-care, disability, permanent impairment, and death benefits. Rates are set by the SFC. The ECAT Fund (formerly FONSAT/Fosyga) covers accidents involving unidentified or uninsured vehicles.
- C. SOAT covers only third-party property damage; personal injuries are paid directly by the health-care system without insurer participation.
- D. Each insurer freely sets its SOAT rates based on its own loss experience.

Question 6: Correct answer: B

In Colombia, SOAT is a **no-fault insurance scheme**: proving the driver's fault is unnecessary; being the victim of a traffic accident is sufficient to access benefits.

- **Coverage:** medical and rehabilitation expenses, temporary disability, permanent impairment, death, and funeral expenses for victims, including vehicle occupants and third parties.
- **Compulsory purchase:** mandatory for every motor vehicle operating in Colombia, regardless of type (car, motorcycle, truck, or public-transport vehicle).
- **Rates:** set by the Financial Superintendence of Colombia (SFC) and differentiated by vehicle type, engine displacement, and use. Insurers cannot price SOAT freely.
- **ECAT Fund:** the Administrator of Catastrophic Traffic Events, formerly FONSAT/Fosyga. It covers victims of accidents caused by uninsured, unidentified, or hit-and-run vehicles. It is financed by a portion of every SOAT premium issued.
- **Actuarial reserves:** SOAT is a relatively short-tail product compared with general liability, but it has high claim frequency, especially for motorcycles. SOAT IBNR requires robust actuarial techniques because of the portfolio's large volume.

7. Liability · P&C · Industrial Fire Calculation · Development tail and IBNR

An actuary compares the IBNR of a General Liability portfolio with that of an Industrial Fire portfolio. Both have the same level of reported losses. The data at 36 months are:

Portfolio	Reported losses at 36 months (COP million)	Age-to-ultimate CDF at 36 months
General Liability	1,200	1.85
Industrial Fire	1,200	1.08

- (a) Calculate the IBNR of each portfolio using Chain Ladder and the undeveloped percentage (α) for each.
- (b) Why is the General Liability development tail much longer than the Industrial Fire tail? State at least four technical reasons.
- (c) What implications does this have for pricing a long-term liability product?

Question 7: Liability IBNR = 1,020 million; Fire IBNR = 96 million

(a) IBNR and undeveloped-percentage calculation

General Liability (CDF = 1.85):

$$\begin{aligned}\text{Ultimate} &= 1,200 \times 1.85 = 2,220 \text{ COP million,} \\ \text{IBNR}_{\text{GL}} &= 2,220 - 1,200 = \mathbf{1,020} \text{ COP million,} \\ q_{\text{GL}} &= 1 - \frac{1}{1.85} \approx 1 - 0.5405 = \mathbf{45.9\%}.\end{aligned}$$

At 36 months, 45.9% remains undeveloped.

Industrial Fire (CDF = 1.08):

$$\begin{aligned}\text{Ultimate} &= 1,200 \times 1.08 = 1,296 \text{ COP million,} \\ \text{IBNR}_{\text{Fire}} &= 1,296 - 1,200 = \mathbf{96} \text{ COP million,} \\ q_{\text{Fire}} &= 1 - \frac{1}{1.08} \approx 1 - 0.9259 = \mathbf{7.4\%}.\end{aligned}$$

At 36 months, 7.4% remains undeveloped.

$$\frac{\text{IBNR}_{\text{GL}}}{\text{IBNR}_{\text{Fire}}} = \frac{1,020}{96} \approx \mathbf{10.6}.$$

General Liability IBNR is approximately **10.6 times** Industrial Fire IBNR.

(b) Reasons why the Liability tail is much longer

- **Delayed claim reporting:** the victim may take years to discover the harm (medical professional liability may involve latent malpractice injuries; product liability may involve defects discovered much later).
- **Prolonged litigation:** civil liability proceedings in Colombia may take 5–10 years. The claim is reported, but the final amount remains unknown until the judgment becomes final.
- **Loss escalation:** fire damage is physical and immediate, often visible on the same day. In liability, non-economic damages, loss of earnings, and medical complications may evolve and increase for years.

- **Long limitation periods:** civil liability actions in Colombia may have limitation periods of 3–10 years depending on the type of harm, allowing very late claims.
- **Occurrence policies:** when coverage is written on an Occurrence basis, the insurer may respond years after the policy expires, generating difficult-to-capture IBNR “outside the triangle.”
- **Fire is physical first-party coverage:** the damage is visible, measurable, and immediate. Settlement occurs within months rather than years.

(c) **Implications for long-term liability pricing**

- The actuary must use explicit **investment rates** to discount future payments when pricing. For long-tail products, investment income compensates for the delay before payment.
- **Loss-inflation assumptions** are critical: judicial and medical escalation over ten years may differ substantially from general inflation.
- The **reserving model** should be more conservative: Bornhuetter–Ferguson or Cape Cod may be preferable to pure Chain Ladder for immature accident years with $q > 40\%$.
- **Run-off reinsurance** or retroactive-cancellation clauses become especially important for controlling long-term exposure.